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Design of Resilient Wide-Area Damping Controllers using a Modified Sine Cosine Algorithm

HENRIQUE R. DE ALMEIDA¹, and MURILO E. C. BENTO¹, (Member, IEEE)

¹Department of Electrical Engineering, Federal University of Rio de Janeiro, Rio de Janeiro, RJ 21941-909, Brazil (e-mails: henrique.res@poli.ufrj.br, murilobento@poli.ufrj.br)

Corresponding author: Murilo E. C. Bento (e-mail: murilobento@poli.ufrj.br).

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ABSTRACT The expansion of power systems due to increased demand and new generation sources has created operational challenges and compromised the continuous supply of electricity. Power system interconnection and operating dynamics can lead to the emergence of oscillation modes with low damping ratios, destabilizing the power system. Recent advances in Synchronized Phasor Measurement Systems have fostered the development of damping controllers called Wide-Area Damping Controllers (WADCs), capable of providing high damping ratios to oscillation modes. However, WADCs are vulnerable to communication failures or cyberattacks such as Denial-of-Service attacks. This paper proposes a Modified Sine Cosine Algorithm for the design of WADCs resilient to communication failures or cyberattacks. Case studies and discussions are conducted in the IEEE 68-bus system using modal analysis and time-domain simulations.

INDEX TERMS Cyberattacks, Cybersecurity, Denial-of-Service Attacks, Low-Frequency Oscillation Modes, Power System Stability, Small-Signal Stability, Smart Grids, Sine Cosine Algorithm, Wide-Area Damping Controllers.

I. INTRODUCTION

THE interconnection of large power systems over the years has brought significant benefits to the continuity of electricity supply to consumption centers. However, challenges have arisen in the operation of large, modern power systems due to various factors such as intermittent power generation, short circuits in transmission lines, sudden load fluctuations, and other factors [1]. Thus, there is a constant search for research and projects by the scientific and industrial community in the development of new tools and equipment that can assist in the safe and desirable operation of modern power systems [1].

During the operation of a power system subject to contingencies, oscillation modes with low damping ratios can negatively affect power system stability [2]. These oscillation modes can be local if they are in the frequency range between 0.8 and 2 Hz and can be inter-area if they are in the frequency range between 0.1 and 0.8 Hz [2]. These low-frequency oscillation (LFOs) modes are the focus of small-signal stability studies of modern power systems, and strategies are studied

to ensure high damping ratios for these modes [2].

Power System Stabilizers (PSSs) have been the primary damping control strategy employed over the years to increase the damping ratios of critical oscillation modes [2]. PSSs are installed in the excitation loops of synchronous generators and provide an additional control signal to Automatic Voltage Regulators (AVRs) [2]. Different PSS design strategies have been developed, including methods based on Linear Matrix Inequalities (LMIs) [3], Optimal Control Theory [4], and metaheuristics [5]. Properly designed PSSs perform well on local modes but have limited or insufficient effect on inter-area oscillation modes because they lack system-wide observability. The interconnection and expansion of large systems have increased the number of inter-area modes with low damping ratios and have required the development of new damping control strategies to assist PSSs in ensuring small-signal stability.

Technological advances and the need for better monitoring systems for large power systems led to the development of Wide-Area Measurement Systems (WAMSs) and the creation

of the Smart Grid concept [6]. WAMSs are equipped with Phasor Measurement Units (PMUs) located at different positions in the system and are capable of measuring three-phase voltages and currents at high sampling rates [6]. This PMU data is time-synchronized thanks to GPS and is sent to the Phasor Data Concentrator (PDC) for possible operation center application [6]. The scientific community has proposed various applications of PMU data in the monitoring [7], control [8], [9], and protection [10] of modern power systems. The reported and proven benefits increasingly motivate the development of new tools based on PMU data.

PMU data has proven useful in damping control design. PMU data can be used to estimate the speed signals of the system's generators and serve as inputs to central damping controllers, also known as Wide-Area Damping Controllers (WADCs). Because WADCs operate with remote signals, greater system observability can be achieved, and inter-area oscillation modes can achieve higher damping ratios than traditional PSSs [11]. In [11], the authors evaluated in a real-time implementation the benefits of WADCs over PSSs and the electromechanical modes are better damped with the use of WADC. Just like PSS design, there are different approaches to WADC design, such as techniques based on LMIs [12], Optimal Control Theory [13], Decision Tree [14] and metaheuristics [15]–[17]. However, there are additional challenges to successful WADC design, such as (i) appropriate selection of the WADC input and output signals; (ii) modeling the time delay in PMU data transmission; and (iii) resilience to communication failures and cyberattacks on data transmission channels.

Techniques based on geometric measures [18], heuristics [19], adaptive signal selection [20] have been developed for the optimal selection of input and output signals for the WADC to be designed. Different time delay models have been developed and studied by researchers. There are fixed time delay models represented by Padé approximations of varying orders [21]–[23]. In this case, the authors consider a maximum time delay limit and, therefore, tolerate a certain variability in the time delay. In [24], [25], the authors use time delay compensators to deal with this in the design of a WADC. In [26], the authors decided to schedule the delay's value for the WADC project. In [27], the author considers time delay as a design variable and seeks to achieve high performance rates for the WADC. In [28], the authors propose to compensate for random time delays by an improved Smith Predictor. In [29], the author decided to determine a time delay margin that data transmission from PMUs should respect. Thus, the scientific community has proposed different tools to select appropriate signals for the design of a WADC and has also proposed different strategies to represent and deal with delays in data transmission from PMUs.

A recent concern is the vulnerability of PMU-equipped systems to cyberattacks. In 2015, a well-designed cyberattack negatively impacted the Ukrainian power system and required the system to adopt improved cyberattack protection measures [30]. Thus, the operation of a WADC is also subject to

various types of cyberattacks that can destabilize the system or compromise small-signal stability. In [31], the authors proposed a method capable of guaranteeing the robustness of the WADC to a single permanent communication failure. In [32], the authors proposed applying redundant signals when a communication channel is experiencing failures. In [33], the authors propose the design of an emergency control scheme to stabilize the system when it is subject to communication failures. In [34], the authors propose to deal with communication failures through a WADC based on detection and correction matrix that changes the matrix gain when a signal is lost. In [35], the author proposes using genetic algorithms and the theory of linear quadratic regulator to treat a communication failure.

In [36], the authors proposed a specific strategy to deal with deception attacks. In [37], [38], the authors proposed different strategies to ensure the desirable operation of a WADC when false data injection attacks occur. There are few studies focusing on the resilient design of WADCs subject to communication failures and cyberattacks. Furthermore, it is difficult to determine the WADC parameters that meet the desired requirements.

This paper proposes the design of a WADC resilient to permanent communication failure by modifying its gains. The project consists of solving a constrained optimization model using a modified sine cosine algorithm. The constraints on the design variables affect the performance of the traditional sine cosine algorithm and motivated the development of the modified sine cosine algorithm. The IEEE 68-bus system was used for modal analysis and time-domain simulation in the case studies. Comparative analyses with other existing WADC design methods are also presented and discussed.

The contribution is presented and discussed in this paper according to the following organization. Section I presents a literature review and the authors' contributions. Section III describes the modeling used for the resilient design of WADCs, including the power system model, the PMU data transmission delay model, the WADC model, and the strategy for resilience to permanent communication failures. Section IV describes the proposed WADC design using the optimization model, the sine cosine algorithm, and the modified sine cosine algorithm. Section V presents and discusses case studies through modal and time-domain analyses. Section VI concludes the paper with key insights and future research lines.

II. TWO-LEVEL CONTROL STRUCTURE

Figure 1 describes the desired control structure for WADC operation in a power system equipped with PMUs. PMU data is transmitted with time delays to the WADC via communication channels. The WADC emits control signals that are sent with transmission delays to the comparator of the synchronous generator AVR, providing an additional control signal. The communication channels are highlighted in red in Figure 1. The authors consider optical fibers as the communication channels because they offer high signal transmission

rates. This control structure is two-level, with a local PSS-type controller and a centralized WADC-type controller sending a control signal to the AVR, totaling two control signals. Typically, the centralized control structure of a WADC with multiple input and multiple output signals is commonly used in the scientific community for the operation of a WADC because it improves the observability of the system and more effectively dampens inter-area oscillation modes where local controllers of the PSS type have more difficulty improving damping rates [39]–[48]. In this research, the PSS and AVR parameters will be fixed, and the control objective is to determine the WADC that meets an optimization model.

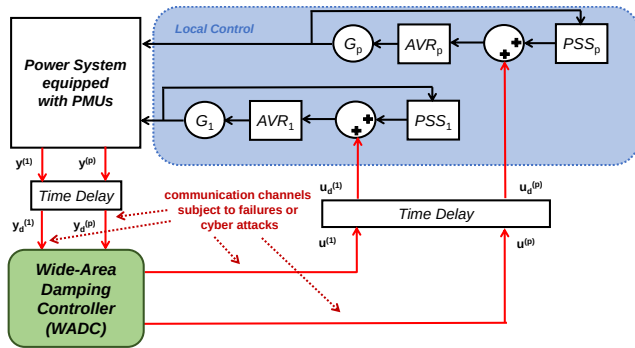


FIGURE 1. Two-level control structure.

III. MODELING

The design of the WADC resilient to communication failures will be performed using linearized models of the system equipped with PMUs, modeling the PMU data transmission delay, and the WADC itself. The performance analysis will consist of evaluating the damping ratios of all eigenvalues of the closed-loop control system and achieving values for these ratios. The use of linearized models is a very common and effective strategy in a wide variety of control projects. Section III-A presents the modeling of power systems with PMU data transmission delays. Section III-B describes the modeling of the fault-resilient WADC to be designed. Section III-C describes the modeling of the closed-loop control system and how the performance requirements will be evaluated. Section III-D presents in detail the strategy for ensuring the WADC's resilience to communication failures.

A. POWER SYSTEM MODEL WITH TIME DELAYS

Today's modern power systems are composed of various equipment, such as synchronous generators in hydroelectric and thermoelectric plants, asynchronous generators in wind farms, direct current and alternating current transmission lines, static and dynamic loads, induction motors, transformers, batteries, reactor and capacitor banks, among other equipment. Accurately modeling the operation of a power system requires the development of a highly detailed set of differential-algebraic equations. In control designs, these

differential-algebraic equations are linearized for each operating point and result in the following state-space model [2]

$$\dot{x} = Ax + Bu \quad (1)$$

$$y = Cx \quad (2)$$

where $x \in \mathbb{R}^n$ represents the system state vector, such as generator angle and speed; $u \in \mathbb{R}^p$ represents the vector of input variables, such as the control signals sent by the WADC; and $y \in \mathbb{R}^p$ represents the vector of output variables, such as the generator speed signals that will be inputs to the WADC. Matrices $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times p}$, and $C \in \mathbb{R}^{p \times n}$ are called state, input, and output matrices, respectively, and are considered fixed and known in the WADC design.

Sending data from PMUs to the WADC input and sending control signals from the WADC to the generators' AVR are subject to time delays. This research employed the following formulation, which consists of the second-order Padé approximation [21]

$$H(s) = \frac{6 - 2Ts}{6 + 4Ts + T^2s^2} \quad (3)$$

where T represents the maximum delay value allowed in data transmission in the WADC communication channels with the system equipped with PMUs.

The time delay model is given by the second-order Padé approximation (3) and represents the maximum time delay allowed for the signal to be effectively applied. The authors consider that there is a buffer at the input and output of the WADC that holds the signal up to a maximum delay limit of T . Thus, when the signal is transmitted with a time less than T , this signal is held up to a time T . Thus, this strategy ensures the system's operational robustness against time delay variability because if the signal is transmitted with a variable delay less than T , the buffer holds this signal until it is actually used.

The transfer function representing the time delay can be formulated in state-space equations using the Jordan canonical representation. Since there are data transmission delays at the input and output of the WADC, there is a state-space model representing the input delay of the WADC (di) given by

$$\dot{x}_{di} = A_{di}x_{di} + B_{di}u_{di} \quad (4)$$

$$y_{di} = C_{di}x_{di} \quad (5)$$

and there is a state-space model representing the output delay of the WADC (do) given by

$$\dot{x}_{do} = A_{do}x_{do} + B_{do}u_{do} \quad (6)$$

$$y_{do} = C_{do}x_{do} \quad (7)$$

The power system state-space model (1)-(2) and time delay models (4)-(7) will be fixed in the resilient WADC design and can thus be grouped together to form the following system

$$\dot{\bar{x}} = \bar{A}\bar{x} + \bar{B}\bar{u} \quad (8)$$

$$\bar{y} = \bar{C}\bar{x} \quad (9)$$

and the matrices $\bar{A}$, $\bar{B}$ and $\bar{C}$ are

$$\bar{A} = \begin{bmatrix} A & BC_{di} & 0 \\ 0 & A_{di} & 0 \\ B_{do}C & 0 & A_{do} \end{bmatrix} \quad (10)$$

$$\bar{B} = \begin{bmatrix} 0 \\ B_{di} \\ 0 \end{bmatrix} \quad (11)$$

$$\bar{C} = [0 \quad 0 \quad C_{do}] \quad (12)$$

In the WADC design, the matrices $\bar{A}$, $\bar{B}$ and $\bar{C}$ are fixed, and a modal analysis is conducted to identify the modes with the lowest damping ratios and which generators will be selected to have their speed measurements estimated and used as WADC input signals. This modal analysis is important to ensure that the WADC increases the damping ratios of the system modes.

B. WADC MODEL

To achieve the benefits of WADC in damping the system's LFO modes, a WADC with multiple inputs and multiple outputs was chosen. The WADC can be formulated and understood as a matrix of transfer functions given by

$$G(s) = [g_{k,m}(s)] = \begin{bmatrix} g_{1,1}(s) & \cdots & g_{1,p}(s) \\ \vdots & \ddots & \vdots \\ g_{p,1}(s) & \cdots & g_{p,p}(s) \end{bmatrix} \quad (13)$$

where each $g_{k,m}(s)$, $k = 1, \dots, p$, $m = 1, \dots, p$ is represented by

$$g_{k,m}(s) = K_{k,m} \cdot \frac{T1_{k,m}s + 1}{T2_{k,m}s + 1} \cdot \frac{T3_{k,m}s + 1}{T4_{k,m}s + 1} \quad (14)$$

where $K_{k,m}$ is the gain of the transfer function that must obey minimum and maximum limits (K_{min} , K_{max}) and $T1_{k,m}$, $T2_{k,m}$, $T3_{k,m}$, $T4_{k,m}$ are time constants that must have values between 0 and 1.

The transfer function matrix $G(s)$ can have its own state-space model if we use the canonical Jordan representation and thus we obtain

$$\dot{x}_c = A_c x_c + B_c u_c \quad (15)$$

$$y_c = C_c x_c + D_c u_c \quad (16)$$

In this research, $K_{k,m}$, $T1_{k,m}$, $T2_{k,m}$, $T3_{k,m}$, $T4_{k,m}$, $k = 1, \dots, p$ and $m = 1, \dots, p$ are design variables.

The model given by (15)-(16) is widely used by researchers in the design of a WADC [39]–[44]. Centralized control structures with multiple inputs and multiple outputs have proven effective in damping inter-area oscillation modes because

they have greater observability of the system. The output control signal of the WADC will be the combination of multiple input signals (speed signals) from different generators in the system.

C. CLOSED-LOOP CONTROL SYSTEM

The successful WADC design of this research must meet certain performance requirements for a closed-loop control system. It is possible to formulate a set of state-space equations that incorporates the time-delayed system modeling (8)-(9) and the WADC modeling (15)-(16). This model is given by

$$\dot{\hat{x}} = \hat{A}\hat{x} \quad (17)$$

where $\hat{x} = [\bar{x}^T \quad x_c^T]^T$ and matrix $\hat{A}$ is

$$\hat{A} = \begin{bmatrix} \bar{A} + \bar{B}D_c\bar{C} & \bar{B}C_c \\ B_c\bar{C} & A_c \end{bmatrix} \quad (18)$$

In this research, one of the performance requirements that must be met by the WADC to be designed is to achieve high damping ratios (ζ) for the closed-loop control system. Thus, if we define $f_{obj}^1()$ as the objective function that calculates the lowest damping ratio ($\zeta_{min}()$) of the closed-loop control system $\hat{A}$ for a given value of the WADC parameters ($K_{k,m}$, $T1_{k,m}$, $T2_{k,m}$, $T3_{k,m}$, $T4_{k,m}$, $k = 1, \dots, p$), then we desire values above a certain minimum damping ratio value (ζ_1) as defined by

$$f_{obj}^1(K_{k,m}, T1_{k,m}, T2_{k,m}, T3_{k,m}, T4_{k,m}) = \zeta_{min}(\hat{A}) \geq \zeta_1 \quad (19)$$

D. WADC RESILIENCE STRATEGY FOR PERMANENT COMMUNICATION FAILURES

The communication failures analyzed in this research will be permanent, meaning one of the WADC's input or output signals will be completely lost, compromising control. In [49], the authors demonstrated that a well-designed Denial-of-Service attack can congest communication channels to the point of permanently losing that communication channel and destabilizing the system. Thus, ensuring WADC resilience to permanent communication loss can prevent the harmful effects of Denial-of-Service attacks.

A permanent loss of communication in one of the WADC input channels can be formulated as zeroing the row of the output matrix $\bar{C}$ corresponding to the lost channel. Now, a permanent loss of communication in one of the WADC output channels can be formulated as zeroing the column of the input matrix $\bar{B}$ corresponding to the lost channel. Thus, in practical terms, the lost signal assumes a zero value in the control action of the closed-loop control system. Since there are p WADC input signals and p WADC output signals, there are p combinations for zeroing matrices $\bar{B}$ and $\bar{C}$. Each of these combinations affects the closed-loop control system, thus altering the eigenvalues and damping ratios. We can then formulate two new matrices for the closed-loop control

system: a matrix corresponding to the loss combinations of the WADC input channels

$$\check{A}^j = \begin{bmatrix} \bar{A} + \bar{B}D_c\bar{C}^j & \bar{B}C_c \\ B_c\bar{C}^j & A_c \end{bmatrix}, \quad j = 1, \dots, p \quad (20)$$

and a matrix corresponding to the loss combinations of the WADC output channels.

$$\tilde{A}^j = \begin{bmatrix} \bar{A} + \bar{B}^j D_c \bar{C} & \bar{B}^j C_c \\ B_c \bar{C} & A_c \end{bmatrix}, \quad j = 1, \dots, p \quad (21)$$

Again, we can ensure that WADC meets performance requirements and is resilient to communication failures by evaluating the eigenvalues and damping rates of the matrices $\check{A}^j$ and $\tilde{A}^j$ defined in (20)-(21) respectively. If we use $f_{obj}^2()$ and $f_{obj}^3()$ as the objective functions that calculate the lowest damping ratio ($\zeta_{min}()$) of the closed-loop control systems $\check{A}^j$ and $\tilde{A}^j$ for the WADC parameters ($K_{k,m}, T1_{k,m}, T2_{k,m}, T3_{k,m}, T4_{k,m}, k = 1, \dots, p$), then we desire values above minimum damping ratio values (ζ_2, ζ_3) as defined by

$$f_{obj}^2(K_{k,m}, T1_{k,m}, T2_{k,m}, T3_{k,m}, T4_{k,m}) = \zeta_{min}(\check{A}) \geq \zeta_2 \quad (22)$$

$$f_{obj}^3(K_{k,m}, T1_{k,m}, T2_{k,m}, T3_{k,m}, T4_{k,m}) = \zeta_{min}(\tilde{A}) \geq \zeta_3 \quad (23)$$

The WADC is robust to multiple operating conditions if a set of linearized matrices of these operating points is considered during the WADC design.

IV. PROPOSED METHOD

The objective of this work is to determine the parameters of a WADC damping controller that provides high damping rates for all modes of the system, even when a communication failure occurs in the channels. These parameters will be determined by solving the optimization model described in Section IV-A. The modified sine cosine algorithm for solving this optimization model is described in Section IV-C, while Section IV-B describes the traditional sine cosine algorithm. The proposed algorithm is presented in Section IV-D.

A. OPTIMIZATION MODEL

The optimization model for this work is described in (24) and is a minimization problem. The vector of variables (x) to be determined for this research optimization problem consists of the WADC parameters ($K_{k,m}, T1_{k,m}, T3_{k,m}$, for $k = 1, \dots, p, m = 1, \dots, p$). The greater the number of input (p) and output (p) signals of the WADC, the greater the number of variables in the problem, and this can make it difficult to achieve good objective function values using the method. It was decided to fix the time constants relative to the poles of the WADC transfer functions to reduce the number of variables. This is an effective strategy for reducing variables, but the choice of poles must be made carefully as it affects the objective function values for the convergence of the proposed method. Usually, the poles of the damping controllers can be set to

values between -10 and -100 and provide good objective function values in the process of searching for the numerator parameters. The objective function ($f_{obj}()$) in (25) to be minimized is multi-objective and has three components. The first component consists of ensuring that the smallest damping ratio of the system operating with the WADC without faults ($\zeta_{min}(\hat{A})$ in (19)) is close to ζ_1 . The second component consists of ensuring that the smallest damping ratio of the system operating with the WADC with a failure in the input channels ($\zeta_{min}(\check{A})$ in (22)) is close to ζ_2 . The third component consists of ensuring that the smallest damping ratio of the system operating with the WADC with a failure in the output channels ($\zeta_{min}(\tilde{A})$ in (23)) is close to ζ_3 . In this minimization problem, the variables $K_{k,m}, T1_{k,m}$ and $T3_{k,m}$ are subject to minimum and maximum limits that must be respected.

$$\begin{aligned} \text{Find} \quad & x = [K_{k,m} \quad T1_{k,m} \quad T3_{k,m}] \\ \text{Minimize} \quad & f_{obj}(x) \\ \text{Subject to} \quad & K_{min} < K_{k,m} < K_{max} \\ & 0 < T1_{k,m} < T_{max} \\ & 0 < T3_{k,m} < T_{max} \end{aligned} \quad (24)$$

$$f_{obj}(x) = (\zeta_{min}(\hat{A}) - \zeta_1)^2 + (\zeta_{min}(\check{A}) - \zeta_2)^2 + (\zeta_{min}(\tilde{A}) - \zeta_3)^2 \quad (25)$$

Various metaheuristics developed by the scientific community are capable of solving this optimization model, such as the Sine Cosine Algorithm [50], Particle Swarm Optimization [51] and Crow Search Algorithm [52], but convergence results can vary and, in some cases, yield unsatisfactory results. Modifications to metaheuristics are common due to the demands of solving complex optimization problems. In this research, a Modified Sine Cosine Algorithm (MSCA) was developed. The following sections describe the traditional SCA and MSCA.

B. SINE COSINE ALGORITHM

The Sine Cosine Algorithm was developed by [50] to solve optimization problems. Over the years, this algorithm has proven effective in solving different problems, as reported in [53]–[57]. The proposed algorithm uses operators based on the sine and cosine functions to explore the exploitation and exploration phases of the search space. Consider a vector of variables X of N_d dimensions ($i = 1, \dots, N_d$) with their values randomly initialized, respecting minimum and maximum limits. In the t -th iteration of the algorithm ($t = 1, \dots, N_T$), this vector is changed according to one of the two equations below.

$$X_i^{t+1} = \begin{cases} X_i^t + r_1 \cdot \sin(r_2) \cdot |r_3 \cdot P_i^t - X_i^t|, & r_4 < 0.5 \\ X_i^t + r_1 \cdot \cos(r_2) \cdot |r_3 \cdot P_i^t - X_i^t|, & r_4 \geq 0.5 \end{cases} \quad (26)$$

where r_1 assumes values according to (27), r_2 is a random value between 0 and $2 \cdot \pi$ from a uniform distribution, and r_3 and r_4 are random values from a uniform distribution between 0 and 1.

$$r_1 = a - t \cdot \frac{a}{N_T} \quad (27)$$

where a is a constant with a usual value of 3 and N_T is the proposed maximum number of iterations [50]. As mentioned, SCA has several applications, but presents some difficulties in finding a desirable solution to the problem under analysis. In many cases, the variables must respect minimum and maximum limits, and the operations in (26) for updating the variable vector may exceed these limits, requiring the variable vector not to be updated. Thus, we may have many cases where the variable vector X does not change throughout these iterations, hindering the process of identifying the best solution to the optimization problem. Furthermore, SCA can provide local solutions far from a desired value due to premature convergence to a local optimum. Thus, modifications to SCA could be beneficial in some optimization problems involving the resilient design of a WADC.

C. MODIFIED SINE COSINE ALGORITHM

SCA may present convergence problems in complex optimization problems. Therefore, modifications were made to its operators described in (26) to achieve better objective function values. In MSCA, the vector of variables can be changed according to one of the three operations described in (28) below

$$X_i^{t+1} = \begin{cases} P_i^t + r_1 \cdot \sin(r_2) \cdot |r_3 \cdot P_i^t - X_i^t|, & r_4 < 0.25 \\ P_i^t + r_1 \cdot \cos(r_2) \cdot |r_3 \cdot P_i^t - X_i^t|, & 0.25 \leq r_4 < 0.5 \\ P_i^t \times r_5, & r_4 \geq 0.5 \end{cases} \quad (28)$$

where r_1 assumes values according to (27), r_2 is a random value between 0 and $2 \cdot \pi$ from a uniform distribution, r_4 are random values from a uniform distribution between 0 and 1. The parameter r_3 will assume a random value from a uniform distribution between 1 and 5 in the first half of the iterations and will assume a random value from a uniform distribution between 0 and 1 in the second half of the iterations. The parameter r_5 will assume a random value from a uniform distribution between 0.8 and 1.2. It is observed that the use of P_i^t increases the elitism of the process of updating the vector of variables X_i^{t+1} . The third operation is not present in the traditional SCA and can bring benefits to the process of minimizing the optimization problem. The first two operations are similar to those presented by the authors [58], [59]. The incorporation of the third operation provides a contribution to the algorithm.

D. PROPOSED ALGORITHM

Based on the description of the proposed optimization model and the Modified Sine Cosine Algorithm, the step-by-step algorithm for obtaining the WADC is described below and the flowchart of the proposed algorithm is presented in Figure 2.

Step 1: Obtain the complete differential algebraic equations of the power system under study and the set of operating

points for which you want to evaluate the system.

Step 2: Linearize the equations around each operating point and obtain the matrices of the linearized model.

Step 3: Apply modal analysis techniques to calculate the eigenvalues of the state matrices and determine the oscillation modes in the frequency range between 0.1 and 2 Hz with the lowest damping ratios.

Step 4: Apply the techniques of geometric measures of controllability and observability to determine the input and output signals of the WADC respectively associated with the modes with the lowest damping ratios.

Step 5: Choose the parameters of the proposed method such as the number of individuals in the population, the number of iterations, among other parameters.

Step 6: Simulate the proposed method until reaching the previously defined maximum iteration limit.

Step 7: When you reach the iteration limit, choose the WADC parameters that provided the best objective function.

Step 8: Evaluate the closed-loop control system with the chosen WADC and check whether the eigenvalues (oscillation modes) present high damping ratio values.

Step 9: Evaluate the closed-loop control system in the time domain with the complete differential algebraic model of the system through the application of contingencies and loss of communication channels.

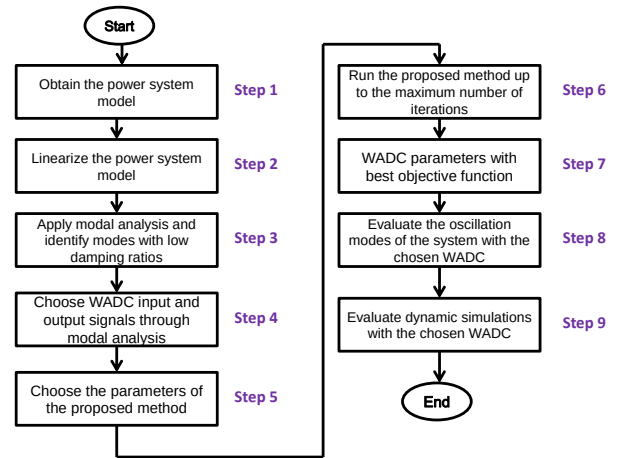


FIGURE 2. Flowchart of the Proposed Method.

V. CASE STUDIES AND DISCUSSION

The small-signal stability studies and evaluation of the proposed method will be performed on the IEEE 68-bus system. This system was developed by [60] for small-signal stability studies and consists of 16 generators and 68 buses. Ref. [60] provides a single operating case (OC0) of this test system. Multiple operating points were also evaluated in this research and consisted of increasing the load level by 1%, 2%, and 3%, obtaining the operating cases OC1, OC2, and OC3, respectively. Table 1 presents the modes (eigenvalues) with

the lowest damping ratios, the frequency, and the damping ratio of these modes for each operating case. There are two pairs of oscillation modes with damping ratios lower than 5% for all cases, and this justifies an appropriate design of the WADC to increase the damping ratios for all these cases.

TABLE 1. LFO modes with lower damping ratios for each operating case

Case	N	Modes (Eigenvalues)	Frequency [Hz]	Damping Ratio [%]
OC0	N1	$-0.16568 \pm j4.89173$	0.77854	3.38505
	N2	$-0.11844 \pm j3.26647$	0.51988	3.62364
OC1	N3	$-0.10587 \pm j3.23828$	0.51539	3.26759
	N4	$-0.16502 \pm j4.88944$	0.77818	3.37304
OC2	N5	$-0.09268 \pm j3.20363$	0.50987	2.89186
	N6	$-0.16432 \pm j4.88547$	0.77755	3.36151
OC3	N7	$-0.07966 \pm j3.16135$	0.50314	2.51887
	N8	$-0.16357 \pm j4.87966$	0.77662	3.35016

This test system features 16 generators, but only generators 1 through 12 are equipped with an AVR. Thus, there are 16 speed signal choices that can be estimated from PMU data and serve as input signals to the WADC. However, there are 12 choices for the WADC output control signals to go to an AVR. Geometric measures of controllability and observability were applied to determine the best WADC signals. Tables 2, 3, 4 and 5 present the geometric measures of controllability for cases OC0, OC1, OC2, and OC3, respectively, for the two modes of each case in Table 1. Generators 5, 9, 10, 11, and 12 presented the highest values of geometric measures of controllability and were thus chosen to receive the WADC control signals in their AVRs. Tables 6, 7, 8 and 9 present the geometric measures of observability for cases OC0, OC1, OC2, and OC3, respectively, for the two modes of each case in Table 1. Generators 12, 13, 14, 15, and 16 presented the highest values of geometric measures of observability and were thus chosen to send the speed signals to the WADC input. Thus, the WADC to be determined can be described as (29). The WADC control action is described by (30), where $\Delta\omega_m$ ($m = 12, 13, 14, 15, 16$) represents the WADC input speed signals and VC_k ($k = 5, 9, 10, 11, 12$) represents the WADC output control signals that will go to the AVR comparator.

$$G(s) = \begin{bmatrix} g_{5,12}(s) & g_{5,13}(s) & g_{5,14}(s) & g_{5,15}(s) & g_{5,16}(s) \\ g_{9,12}(s) & g_{9,13}(s) & g_{9,14}(s) & g_{9,15}(s) & g_{9,16}(s) \\ g_{10,12}(s) & g_{10,13}(s) & g_{10,14}(s) & g_{10,15}(s) & g_{10,16}(s) \\ g_{11,12}(s) & g_{11,13}(s) & g_{11,14}(s) & g_{11,15}(s) & g_{11,16}(s) \\ g_{12,12}(s) & g_{12,13}(s) & g_{12,14}(s) & g_{12,15}(s) & g_{12,16}(s) \end{bmatrix} \quad (29)$$

$$\begin{bmatrix} VC_5 \\ VC_9 \\ VC_{10} \\ VC_{11} \\ VC_{12} \end{bmatrix} = G(s) \cdot \begin{bmatrix} \Delta\omega_{12} \\ \Delta\omega_{13} \\ \Delta\omega_{14} \\ \Delta\omega_{15} \\ \Delta\omega_{16} \end{bmatrix} \quad (30)$$

TABLE 2. Controllability factors of N1 and N2 for the OC0 case.

Generator	Controllability Factors		Avaliable?
	N1	N2	
1	0.1828	0.3742	Yes
2	0.1705	0.3611	Yes
3	0.2409	0.4987	Yes
4	0.2588	0.5456	Yes
5	0.2332	0.4812	Yes
6	0.2228	0.4469	Yes
7	0.2518	0.5250	Yes
8	0.1196	0.2275	Yes
9	0.8623	1.0000	Yes
10	0.3655	0.2116	Yes
11	0.6973	0.6118	Yes
12	0.0783	0.7467	Yes
13	0.0597	0.2286	No
14	1.0000	0.3723	No
15	0.8935	0.0790	No
16	0.4959	0.3559	No

TABLE 3. Controllability factors of N3 and N4 for the OC1 case.

Generator	Controllability Factors		Avaliable?
	N3	N4	
1	0.3773	0.1911	Yes
2	0.3606	0.1753	Yes
3	0.4973	0.2464	Yes
4	0.5460	0.2631	Yes
5	0.4823	0.2374	Yes
6	0.4477	0.2265	Yes
7	0.5253	0.2561	Yes
8	0.2293	0.1230	Yes
9	1.0000	0.8812	Yes
10	0.2227	0.3830	Yes
11	0.6958	0.7461	Yes
12	0.8184	0.1472	Yes
13	0.2469	0.0411	No
14	0.5249	1.0000	No
15	0.1021	0.9022	No
16	0.5105	0.5067	No

The proposed method was evaluated comparatively with the traditional SCA, the Particle Swarm Optimization (PSO) [51] and Crow Search Algorithm (CSA) [52]. The parameters of the proposed algorithm were $N_p = 20$ and $N_e = 500$. The parameters of the SCA were $N_p = 20$ and $N_e = 500$. A first comparative analysis consisted of evaluating the convergence results for 100 simulations of four methods. This measure is necessary because each execution of both methods results in different objective function values due to random conditions and other operators with random values. Table 10 presents the minimum, mean, maximum, and standard deviation values of the objective function (f_{obj}). These results allow the following

TABLE 4. Controllability factors of N5 and N6 for the OC2 case.

Generator	Controllability Factors		Avaliable?
	N5	N6	
1	0.3782	0.1992	Yes
2	0.3538	0.1798	Yes
3	0.4879	0.2515	Yes
4	0.5449	0.2667	Yes
5	0.4827	0.2410	Yes
6	0.4470	0.2295	Yes
7	0.5237	0.2598	Yes
8	0.2306	0.1261	Yes
9	1.0000	0.8969	Yes
10	0.2466	0.3996	Yes
11	0.8916	0.7938	Yes
12	0.9711	0.2178	Yes
13	0.2839	0.0284	No
14	0.8604	1.0000	No
15	0.1550	0.9114	No
16	0.8484	0.5150	No

TABLE 5. Controllability factors of N7 and N8 for the OC3 case.

Generator	Controllability Factors		Avaliable?
	N7	N8	
1	0.1756	0.2069	Yes
2	0.1445	0.1841	Yes
3	0.2019	0.2560	Yes
4	0.2532	0.2695	Yes
5	0.2263	0.2439	Yes
6	0.2071	0.2318	Yes
7	0.2420	0.2625	Yes
8	0.1099	0.1289	Yes
9	0.4816	0.9089	Yes
10	0.1687	0.4152	Yes
11	0.8012	0.8402	Yes
12	0.7458	0.2897	Yes
13	0.2004	0.0322	No
14	1.0000	1.0000	No
15	0.1704	0.9217	No
16	0.9956	0.5208	No

discussions:

- The lowest objective function values were achieved by the proposed method (MSCA). The lowest value of 100 MSCA simulations was 6.27727×10^{-8} . The SCA, PSO and CSA methods converged with high values.
- The standard deviation for 100 MSCA simulations was the smallest among the four methods. This indicates that random initialization of the WADC variables does not hinder the convergence of the MSCA at smaller values.
- The proposed MSCA method showed superior performance than the traditional SCA in all 100 simulations performed for this case study.

TABLE 6. Observability factors of N1 and N2 for the OC0 case.

Generator	Observability Factors		Avaliable?
	N1	N2	
1	0.0417	0.2227	Yes
2	0.0287	0.2965	Yes
3	0.0302	0.3046	Yes
4	0.0388	0.3289	Yes
5	0.0442	0.3489	Yes
6	0.0462	0.3439	Yes
7	0.0439	0.3352	Yes
8	0.0317	0.2459	Yes
9	0.0601	0.2829	Yes
10	0.0692	0.2026	Yes
11	0.0295	0.2492	Yes
12	0.0450	0.3666	Yes
13	0.1269	0.5305	Yes
14	0.6309	1.0000	Yes
15	1.0000	0.2213	Yes
16	0.3246	0.9208	Yes

TABLE 7. Observability factors of N3 and N4 for the OC1 case.

Generator	Observability Factors		Avaliable?
	N3	N4	
1	0.2126	0.0434	Yes
2	0.2789	0.0299	Yes
3	0.2866	0.0313	Yes
4	0.3077	0.0402	Yes
5	0.3261	0.0457	Yes
6	0.3214	0.0479	Yes
7	0.3133	0.0455	Yes
8	0.2335	0.0327	Yes
9	0.2694	0.0626	Yes
10	0.1931	0.0705	Yes
11	0.2352	0.0310	Yes
12	0.3460	0.0441	Yes
13	0.5013	0.1265	Yes
14	1.0000	0.6320	Yes
15	0.2315	1.0000	Yes
16	0.8894	0.3123	Yes

Linearized models were applied to the resilient design of a WADC, but power systems are nonlinear dynamic systems. Therefore, the WADC must be evaluated in a nonlinear dynamic system to assess small-signal stability. Four WADCs were evaluated: the WADC from the proposed method (MSCA), which provided the lowest objective function, the WADC from the SCA-based method, which provided the lowest objective function, the WADC from the PSO-based method, which provided the lowest objective function and the WADC from the CSA-based method, which provided the lowest objective function. The parameters of the WADCs from the CSA, PSO, SCA and MSCA methods are detailed

TABLE 8. Observability factors of N5 and N6 for the OC2 case.

Generator	Observability Factors		Avaliable?
	N5	N6	
1	0.1996	0.0450	Yes
2	0.2578	0.0311	Yes
3	0.2648	0.0324	Yes
4	0.2829	0.0415	Yes
5	0.2995	0.0472	Yes
6	0.2952	0.0496	Yes
7	0.2878	0.0471	Yes
8	0.2182	0.0337	Yes
9	0.2520	0.0649	Yes
10	0.1805	0.0716	Yes
11	0.2181	0.0325	Yes
12	0.3207	0.0430	Yes
13	0.4651	0.1254	Yes
14	1.0000	0.6335	Yes
15	0.2456	1.0000	Yes
16	0.8744	0.2992	Yes

TABLE 9. Observability factors of N7 and N8 for the OC3 case.

Generator	Observability Factors		Avaliable?
	N7	N8	
1	0.1839	0.0465	Yes
2	0.2332	0.0323	Yes
3	0.2395	0.0335	Yes
4	0.2550	0.0427	Yes
5	0.2695	0.0486	Yes
6	0.2656	0.0511	Yes
7	0.2591	0.0487	Yes
8	0.1999	0.0346	Yes
9	0.2306	0.0671	Yes
10	0.1648	0.0726	Yes
11	0.1981	0.0338	Yes
12	0.2909	0.0418	Yes
13	0.4221	0.1236	Yes
14	1.0000	0.6351	Yes
15	0.2644	1.0000	Yes
16	0.8449	0.2852	Yes

TABLE 10. Results of the objective function for 100 simulations

Method	Minimum [$\times 10^{-3}$]	Average [$\times 10^{-3}$]	Maximum [$\times 10^{-3}$]	Standard Deviation [$\times 10^{-3}$]
MSCA	0.00006	0.26876	0.59030	0.17552
SCA	2.16459	3.58608	4.49639	0.54516
PSO	3.07061	5.02325	11.25148	1.65218
CSA	5.47294	9.33360	13.55388	1.99828

in Tables 11, 12, 13 and 14, respectively. Tables 15, 16, 17 and 18 present the modes with the lowest damping ratios for each operating case, considering the loss or absence of each communication channel for the WADC-CSA, WADC-PSO, WADC-SCA and WADC-MSCA, respectively. These results allow the following discussions:

- As expected, communication channel losses affect the oscillation modes of the closed-loop control system and may result in cases with low damping ratios.
- The OC0, OC1, OC2, and OC3 cases with the WADC designed by the proposed method (WADC-MSCA) provided high damping ratio values. The cases operating without signal loss provided damping ratios above 11%. The cases with a communication failure provided damping ratios close to 6%.
- The OC0, OC1, OC2, and OC3 cases with WADCs designed using the SCA, PSO, and CSA methods provided damping ratios well below 5% for some communication channel loss scenarios. Low damping ratios can compromise the small-signal stability of modern power systems equipped with PMUs.

TABLE 11. WADC parameters obtained by CSA (WADC-CSA)

$g_{k,m}$	$K_{k,m}$	$T1_{k,m}$	$T3_{k,m}$
$g_{5,12}$	8.1608	0.7341	0.7522
$g_{5,13}$	6.0291	0.4545	0.5978
$g_{5,14}$	22.2333	0.1589	0.1540
$g_{5,15}$	4.6178	0.4554	0.8094
$g_{5,16}$	-0.3870	0.8156	0.0862
$g_{9,12}$	-2.6882	0.0480	0.4119
$g_{9,13}$	-3.4752	0.1900	0.0932
$g_{9,14}$	21.1290	0.5667	0.5068
$g_{9,15}$	12.3630	0.6009	0.3730
$g_{9,16}$	12.2490	0.3992	0.8508
$g_{10,12}$	-21.6633	0.1674	0.3484
$g_{10,13}$	10.2184	0.5057	0.1549
$g_{10,14}$	10.5147	0.2341	0.8204
$g_{10,15}$	16.8301	0.5655	0.1546
$g_{10,16}$	22.6884	0.7453	0.6549
$g_{11,12}$	9.9626	0.6331	0.8079
$g_{11,13}$	9.3128	0.8716	0.3551
$g_{11,14}$	18.6782	0.3606	0.1227
$g_{11,15}$	-1.0566	0.7595	0.7972
$g_{11,16}$	20.2365	0.3721	0.7781
$g_{12,12}$	-3.6470	0.1405	0.3608
$g_{12,13}$	4.2862	0.6221	0.0555
$g_{12,14}$	27.6971	0.0982	0.9287
$g_{12,15}$	4.4519	0.1462	0.6246
$g_{12,16}$	6.7591	0.4122	0.5796

TABLE 12. WADC parameters obtained by PSO (WADC-PSO)

$g_{k,m}$	$K_{k,m}$	$T1_{k,m}$	$T3_{k,m}$
$g_{5,12}$	-26.2628	0.3735	0.3593
$g_{5,13}$	-0.5885	0.7180	0.3580
$g_{5,14}$	15.0761	0.4641	0.1951
$g_{5,15}$	23.3223	0.3551	0.0043
$g_{5,16}$	6.7960	0.1192	0.0968
$g_{9,12}$	6.6124	0.4974	0.1398
$g_{9,13}$	-22.4468	0.8065	0.1205
$g_{9,14}$	19.7631	0.3142	0.9901
$g_{9,15}$	-14.8963	0.0582	0.0505
$g_{9,16}$	20.7780	0.0624	0.0965
$g_{10,12}$	0.1642	0.5812	0.8463
$g_{10,13}$	35.6741	0.3503	0.6671
$g_{10,14}$	11.6919	0.7879	0.2779
$g_{10,15}$	32.1682	0.2007	0.2324
$g_{10,16}$	-15.5324	0.1946	0.9926
$g_{11,12}$	27.3249	0.7159	0.2199
$g_{11,13}$	25.1624	0.7296	0.1950
$g_{11,14}$	18.4480	0.9251	0.1934
$g_{11,15}$	-16.8000	0.6318	0.3834
$g_{11,16}$	24.2230	0.5415	0.7077
$g_{12,12}$	1.0658	0.6124	0.9062
$g_{12,13}$	-0.3326	0.2942	0.6805
$g_{12,14}$	8.2164	0.2746	0.4235
$g_{12,15}$	13.1086	0.4567	0.2192
$g_{12,16}$	20.8588	0.0438	0.0656

TABLE 13. WADC parameters obtained by SCA (WADC-SCA)

$g_{k,m}$	$K_{k,m}$	$T1_{k,m}$	$T3_{k,m}$
$g_{5,12}$	0.1457	0.0744	0.1191
$g_{5,13}$	-35.3580	0.0006	0.0004
$g_{5,14}$	0.0572	0.0087	0.0084
$g_{5,15}$	0.0565	0.0005	0.0017
$g_{5,16}$	-0.0138	0.0025	0.0020
$g_{9,12}$	-1.3840	0.2380	0.1912
$g_{9,13}$	-21.3255	0.0010	0.0002
$g_{9,14}$	22.3701	0.5843	0.6372
$g_{9,15}$	23.3636	0.0542	0.0439
$g_{9,16}$	-0.3313	0.0012	0.0014
$g_{10,12}$	-9.5584	0.0258	0.0258
$g_{10,13}$	0.2284	0.0151	0.0306
$g_{10,14}$	1.7703	0.0337	0.0427
$g_{10,15}$	20.6763	0.5511	0.4965
$g_{10,16}$	0.3737	0.0151	0.0144
$g_{11,12}$	8.3113	0.5348	0.5856
$g_{11,13}$	-0.2002	0.0305	0.0321
$g_{11,14}$	-1.7802	0.0138	0.0133
$g_{11,15}$	0.0525	0.0052	0.0052
$g_{11,16}$	35.8125	0.8888	0.8672
$g_{12,12}$	-1.0968	0.0033	0.0048
$g_{12,13}$	1.4190	0.0581	0.0405
$g_{12,14}$	-0.3296	0.0011	0.0002
$g_{12,15}$	0.4321	0.0739	0.0275
$g_{12,16}$	0.2172	0.0078	0.0066

TABLE 14. WADC parameters obtained by MSCA (WADC-MSCA)

$g_{k,m}$	$K_{k,m}$	$T1_{k,m}$	$T3_{k,m}$
$g_{5,12}$	0.6917	0.2070	0.6374
$g_{5,13}$	0.3451	0.0412	0.0458
$g_{5,14}$	-22.9045	0.2115	0.0710
$g_{5,15}$	39.3769	0.3628	0.9282
$g_{5,16}$	-15.1473	0.8793	0.9636
$g_{9,12}$	-10.6465	0.2921	0.1217
$g_{9,13}$	-14.0170	0.4094	0.0352
$g_{9,14}$	25.0269	0.3305	0.9151
$g_{9,15}$	-3.0845	0.8186	0.6706
$g_{9,16}$	-8.4239	0.0021	0.0125
$g_{10,12}$	16.4943	0.0255	0.0090
$g_{10,13}$	29.6551	0.6554	0.4777
$g_{10,14}$	0.0020	0.3281	0.4046
$g_{10,15}$	0.5781	0.3910	0.0465
$g_{10,16}$	-6.5500	0.6853	0.3645
$g_{11,12}$	21.4418	0.5398	0.4448
$g_{11,13}$	-0.0368	0.5863	0.3357
$g_{11,14}$	-20.6845	0.5211	0.9214
$g_{11,15}$	-2.3703	0.8701	0.8586
$g_{11,16}$	36.1827	0.7797	0.9880
$g_{12,12}$	-19.8851	0.0251	0.0266
$g_{12,13}$	1.8687	0.0113	0.0280
$g_{12,14}$	-1.2965	0.3768	0.0442
$g_{12,15}$	16.6054	0.0493	0.1595
$g_{12,16}$	1.2540	0.3666	0.5538

TABLE 15. LFO modes with lower damping ratios for each operating case with WADC-CSA

Case	Signal Loss	Modes (Eigenvalues)	Frequency [Hz]	Damping Ratio [%]
OC0	None	$-0.2290 \pm 5.0332i$	0.8011	4.5450
	VC ₅	$-0.1873 \pm 5.0412i$	0.8023	3.7123
	VC ₉	$-0.1961 \pm 4.9348i$	0.7854	3.9707
	VC ₁₀	$-0.1988 \pm 4.9593i$	0.7893	4.0046
	VC ₁₁	$-0.1639 \pm 3.7926i$	0.6036	4.3172
	VC ₁₂	$-0.2047 \pm 3.3568i$	0.5343	6.0872
	$\Delta\omega_{12}$	$-0.3833 \pm 5.1634i$	0.8218	7.4026
	$\Delta\omega_{13}$	$-0.1975 \pm 4.9650i$	0.7902	3.9751
	$\Delta\omega_{14}$	$-0.0842 \pm 3.5190i$	0.5601	2.3934
	$\Delta\omega_{15}$	$-0.1814 \pm 5.1247i$	0.8156	3.5385
	$\Delta\omega_{16}$	$-0.1803 \pm 4.9040i$	0.7805	3.6731
OC1	None	$-0.2295 \pm 5.0459i$	0.8031	4.5429
	VC ₅	$-0.1865 \pm 5.0525i$	0.8041	3.6885
	VC ₉	$-0.1969 \pm 4.9399i$	0.7862	3.9834
	VC ₁₀	$-0.1987 \pm 4.9658i$	0.7903	3.9972
	VC ₁₁	$-0.1419 \pm 3.0805i$	0.4903	4.6029
	VC ₁₂	$-0.1876 \pm 3.3006i$	0.5253	5.6745
	$\Delta\omega_{12}$	$-0.3964 \pm 5.2144i$	0.8299	7.5799
	$\Delta\omega_{13}$	$-0.1967 \pm 4.9702i$	0.7910	3.9545
	$\Delta\omega_{14}$	$-0.1008 \pm 3.4291i$	0.5458	2.9368
	$\Delta\omega_{15}$	$-0.1770 \pm 5.1376i$	0.8177	3.4430
	$\Delta\omega_{16}$	$-0.1790 \pm 4.9061i$	0.7808	3.6454
OC2	None	$-0.2298 \pm 5.0566i$	0.8048	4.5408
	VC ₅	$-0.1862 \pm 5.0619i$	0.8056	3.6757
	VC ₉	$-0.1978 \pm 4.9433i$	0.7868	3.9979
	VC ₁₀	$-0.1986 \pm 4.9706i$	0.7911	3.9916
	VC ₁₁	$-0.1184 \pm 3.0880i$	0.4915	3.8309
	VC ₁₂	$-0.1679 \pm 3.2419i$	0.5160	5.1733
	$\Delta\omega_{12}$	$-0.3960 \pm 5.2612i$	0.8373	7.5052
	$\Delta\omega_{13}$	$-0.1958 \pm 4.9734i$	0.7915	3.9348
	$\Delta\omega_{14}$	$-0.1051 \pm 3.3349i$	0.5308	3.1514
	$\Delta\omega_{15}$	$-0.1731 \pm 5.1479i$	0.8193	3.3612
	$\Delta\omega_{16}$	$-0.1776 \pm 4.9067i$	0.7809	3.6172
OC3	None	$-0.2304 \pm 5.0648i$	0.8061	4.5437
	VC ₅	$-0.1866 \pm 5.0692i$	0.8068	3.6785
	VC ₉	$-0.1987 \pm 4.9449i$	0.7870	4.0155
	VC ₁₀	$-0.1986 \pm 4.9735i$	0.7916	3.9898
	VC ₁₁	$-0.0915 \pm 3.0861i$	0.4912	2.9642
	VC ₁₂	$-0.1464 \pm 3.1808i$	0.5062	4.5986
	$\Delta\omega_{12}$	$-0.1748 \pm 3.1240i$	0.4972	5.5879
	$\Delta\omega_{13}$	$-0.1950 \pm 4.9744i$	0.7917	3.9172
	$\Delta\omega_{14}$	$-0.1007 \pm 3.2374i$	0.5152	3.1076
	$\Delta\omega_{15}$	$-0.1701 \pm 5.1552i$	0.8205	3.2971
	$\Delta\omega_{16}$	$-0.1762 \pm 4.9059i$	0.7808	3.5890

TABLE 16. LFO modes with lower damping ratios for each operating case WADC-PSO

Case	Signal Loss	Modes (Eigenvalues)	Frequency [Hz]	Damping Ratio [%]
OC0	None	$-0.2021 \pm 3.1948i$	0.5085	6.3129
	VC ₅	$-0.1438 \pm 3.1748i$	0.5053	4.5238
	VC ₉	$-0.1419 \pm 3.3048i$	0.5260	4.2894
	VC ₁₀	$-0.2505 \pm 5.1233i$	0.8154	4.8832
	VC ₁₁	$-0.0916 \pm 2.9276i$	0.4659	3.1260
	VC ₁₂	$-0.3131 \pm 6.8000i$	1.0823	4.5991
	$\Delta\omega_{12}$	$-0.1346 \pm 3.0605i$	0.4871	4.3949
	$\Delta\omega_{13}$	$-0.1985 \pm 5.1197i$	0.8148	3.8745
	$\Delta\omega_{14}$	$-0.1151 \pm 3.3356i$	0.5309	3.4471
	$\Delta\omega_{15}$	$-0.1915 \pm 3.1630i$	0.5034	6.0418
	$\Delta\omega_{16}$	$-0.2784 \pm 5.1419i$	0.8184	5.4066
OC1	None	$-0.2052 \pm 3.1925i$	0.5081	6.4155
	VC ₅	$-0.1614 \pm 3.1605i$	0.5030	5.1009
	VC ₉	$-0.1287 \pm 3.2699i$	0.5204	3.9337
	VC ₁₀	$-0.2558 \pm 5.1339i$	0.8171	4.9769
	VC ₁₁	$-0.1915 \pm 4.9469i$	0.7873	3.8682
	VC ₁₂	$-0.3149 \pm 6.8026i$	1.0827	4.6243
	$\Delta\omega_{12}$	$-0.1579 \pm 3.0724i$	0.4890	5.1329
	$\Delta\omega_{13}$	$-0.2007 \pm 5.1246i$	0.8156	3.9129
	$\Delta\omega_{14}$	$-0.1135 \pm 3.2889i$	0.5234	3.4476
	$\Delta\omega_{15}$	$-0.3162 \pm 5.0876i$	0.8097	6.2030
	$\Delta\omega_{16}$	$-0.2821 \pm 5.1519i$	0.8199	5.4681
OC2	None	$-0.2051 \pm 3.1847i$	0.5069	6.4273
	VC ₅	$-0.1774 \pm 3.1413i$	0.5000	5.6392
	VC ₉	$-0.1137 \pm 3.2288i$	0.5139	3.5201
	VC ₁₀	$-0.2614 \pm 5.1419i$	0.8184	5.0772
	VC ₁₁	$-0.1914 \pm 4.9428i$	0.7867	3.8704
	VC ₁₂	$-0.3160 \pm 6.8062i$	1.0832	4.6381
	$\Delta\omega_{12}$	$-0.1806 \pm 3.0845i$	0.4909	5.8448
	$\Delta\omega_{13}$	$-0.2030 \pm 5.1270i$	0.8160	3.9573
	$\Delta\omega_{14}$	$-0.1097 \pm 3.2349i$	0.5148	3.3883
	$\Delta\omega_{15}$	$-0.3252 \pm 5.0949i$	0.8109	6.3705
	$\Delta\omega_{16}$	$-0.2864 \pm 5.1596i$	0.8212	5.5427
OC3	None	$-0.2007 \pm 3.1693i$	0.5044	6.3215
	VC ₅	$-0.2311 \pm 3.8455i$	0.6120	5.9981
	VC ₉	$-0.0969 \pm 3.1809i$	0.5063	3.0441
	VC ₁₀	$-0.2674 \pm 5.1471i$	0.8192	5.1875
	VC ₁₁	$-0.1913 \pm 4.9364i$	0.7856	3.8732
	VC ₁₂	$-0.3162 \pm 6.8111i$	1.0840	4.6381
	$\Delta\omega_{12}$	$-0.3847 \pm 6.0878i$	0.9689	6.3062
	$\Delta\omega_{13}$	$-0.2057 \pm 5.1264i$	0.8159	4.0088
	$\Delta\omega_{14}$	$-0.1037 \pm 3.1727i$	0.5049	3.2660
	$\Delta\omega_{15}$	$-0.2051 \pm 3.1628i$	0.5034	6.4699
	$\Delta\omega_{16}$	$-0.2915 \pm 5.1651i$	0.8220	5.6351

TABLE 17. LFO modes with lower damping ratios for each operating case WADC-SCA

Case	Signal Loss	Modes (Eigenvalues)	Frequency [Hz]	Damping Ratio [%]
OC0	None	$-0.3995 \pm 5.0742i$	0.8076	7.8485
	VC ₅	$-0.4173 \pm 5.0419i$	0.8024	8.2494
	VC ₉	$-0.1073 \pm 3.7927i$	0.6036	2.8268
	VC ₁₀	$-0.4452 \pm 5.1473i$	0.8192	8.6178
	VC ₁₁	$-0.0806 \pm 3.1256i$	0.4975	2.5775
	VC ₁₂	$-0.4013 \pm 5.0763i$	0.8079	7.8806
	$\Delta\omega_{12}$	$-0.3702 \pm 5.0217i$	0.7992	7.3513
	$\Delta\omega_{13}$	$-0.2488 \pm 3.6976i$	0.5885	6.7144
	$\Delta\omega_{14}$	$-0.2904 \pm 4.9100i$	0.7814	5.9036
	$\Delta\omega_{15}$	$-0.4137 \pm 5.1320i$	0.8168	8.0343
	$\Delta\omega_{16}$	$-0.0910 \pm 3.1513i$	0.5015	2.8878
OC1	None	$-0.4167 \pm 5.0763i$	0.8079	8.1811
	VC ₅	$-0.4374 \pm 5.0399i$	0.8021	8.6471
	VC ₉	$-0.1982 \pm 3.7792i$	0.6015	5.2380
	VC ₁₀	$-0.4640 \pm 5.1582i$	0.8210	8.9587
	VC ₁₁	$-0.0826 \pm 3.1307i$	0.4983	2.6380
	VC ₁₂	$-0.4190 \pm 5.0785i$	0.8083	8.2234
	$\Delta\omega_{12}$	$-0.3805 \pm 5.0169i$	0.7985	7.5617
	$\Delta\omega_{13}$	$-0.3670 \pm 5.0443i$	0.8028	7.2561
	$\Delta\omega_{14}$	$-0.2984 \pm 4.8998i$	0.7798	6.0798
	$\Delta\omega_{15}$	$-0.4319 \pm 5.1390i$	0.8179	8.3755
	$\Delta\omega_{16}$	$-0.0937 \pm 3.1516i$	0.5016	2.9720
OC2	None	$-0.4335 \pm 5.0741i$	0.8076	8.5129
	VC ₅	$-0.4573 \pm 5.0326i$	0.8010	9.0496
	VC ₉	$-0.2704 \pm 4.8991i$	0.7797	5.5112
	VC ₁₀	$-0.4813 \pm 5.1654i$	0.8221	9.2782
	VC ₁₁	$-0.0833 \pm 3.1325i$	0.4986	2.6576
	VC ₁₂	$-0.4365 \pm 5.0762i$	0.8079	8.5677
	$\Delta\omega_{12}$	$-0.3887 \pm 5.0084i$	0.7971	7.7368
	$\Delta\omega_{13}$	$-0.3794 \pm 5.0388i$	0.8019	7.5085
	$\Delta\omega_{14}$	$-0.3043 \pm 4.8856i$	0.7776	6.2164
	$\Delta\omega_{15}$	$-0.4496 \pm 5.1420i$	0.8184	8.7100
	$\Delta\omega_{16}$	$-0.0955 \pm 3.1480i$	0.5010	3.0316
OC3	None	$-0.2423 \pm 3.0702i$	0.4886	7.8669
	VC ₅	$-0.2506 \pm 3.0643i$	0.4877	8.1507
	VC ₉	$-0.2728 \pm 4.8846i$	0.7774	5.5755
	VC ₁₀	$-0.2439 \pm 3.0750i$	0.4894	7.9082
	VC ₁₁	$-0.0827 \pm 3.1296i$	0.4981	2.6407
	VC ₁₂	$-0.2431 \pm 3.0705i$	0.4887	7.8917
	$\Delta\omega_{12}$	$-0.2308 \pm 3.0478i$	0.4851	7.5511
	$\Delta\omega_{13}$	$-0.3902 \pm 5.0281i$	0.8003	7.7381
	$\Delta\omega_{14}$	$-0.3069 \pm 4.8676i$	0.7747	6.2934
	$\Delta\omega_{15}$	$-0.2423 \pm 3.0732i$	0.4891	7.8607
	$\Delta\omega_{16}$	$-0.0965 \pm 3.1393i$	0.4996	3.0715

TABLE 18. LFO modes with lower damping ratios for each operating case WADC-MSCA

Case	Signal Loss	Modes (Eigenvalues)	Frequency [Hz]	Damping Ratio [%]
OC0	None	$-0.4591 \pm 3.6619i$	0.5828	12.4408
	VC ₅	$-0.3464 \pm 3.5528i$	0.5654	9.7028
	VC ₉	$-0.3059 \pm 4.0191i$	0.6397	7.5902
	VC ₁₀	$-0.3687 \pm 5.0894i$	0.8100	7.2256
	VC ₁₁	$-0.1841 \pm 2.9054i$	0.4624	6.3232
	VC ₁₂	$-0.6925 \pm 7.1849i$	1.1435	9.5942
	$\Delta\omega_{12}$	$-0.4440 \pm 6.0197i$	0.9581	7.3565
	$\Delta\omega_{13}$	$-0.3166 \pm 5.1339i$	0.8171	6.1547
	$\Delta\omega_{14}$	$-0.2781 \pm 3.6125i$	0.5749	7.6750
	$\Delta\omega_{15}$	$-0.3800 \pm 4.9421i$	0.7866	7.6666
	$\Delta\omega_{16}$	$-0.2128 \pm 3.2603i$	0.5189	6.5132
OC1	None	$-0.6692 \pm 5.5752i$	0.8873	11.9184
	VC ₅	$-0.3712 \pm 3.4246i$	0.5450	10.7748
	VC ₉	$-0.4681 \pm 4.7785i$	0.7605	9.7490
	VC ₁₀	$-0.3771 \pm 5.0970i$	0.8112	7.3782
	VC ₁₁	$-0.2275 \pm 2.9723i$	0.4731	7.6307
	VC ₁₂	$-0.6819 \pm 7.2001i$	1.1459	9.4279
	$\Delta\omega_{12}$	$-0.4432 \pm 6.0166i$	0.9576	7.3470
	$\Delta\omega_{13}$	$-0.3195 \pm 5.1401i$	0.8181	6.2046
	$\Delta\omega_{14}$	$-0.3093 \pm 3.4814i$	0.5541	8.8505
	$\Delta\omega_{15}$	$-0.3829 \pm 4.9261i$	0.7840	7.7497
	$\Delta\omega_{16}$	$-0.2104 \pm 3.2443i$	0.5163	6.4723
OC2	None	$-0.6426 \pm 5.5661i$	0.8859	11.4691
	VC ₅	$-0.3525 \pm 3.2846i$	0.5228	10.6718
	VC ₉	$-0.4671 \pm 4.6827i$	0.7453	9.9261
	VC ₁₀	$-0.3849 \pm 5.1013i$	0.8119	7.5232
	VC ₁₁	$-0.2377 \pm 3.0696i$	0.4885	7.7201
	VC ₁₂	$-0.6703 \pm 7.2181i$	1.1488	9.2470
	$\Delta\omega_{12}$	$-0.4476 \pm 6.0128i$	0.9570	7.4237
	$\Delta\omega_{13}$	$-0.3223 \pm 5.1425i$	0.8184	6.2543
	$\Delta\omega_{14}$	$-0.2874 \pm 3.3270i$	0.5295	8.6054
	$\Delta\omega_{15}$	$-0.3803 \pm 4.9078i$	0.7811	7.7262
	$\Delta\omega_{16}$	$-0.2067 \pm 3.2208i$	0.5126	6.4055
OC3	None	$-0.3552 \pm 3.2030i$	0.5098	11.0207
	VC ₅	$-0.3023 \pm 3.1608i$	0.5030	9.5212
	VC ₉	$-0.3766 \pm 4.6120i$	0.7340	8.1380
	VC ₁₀	$-0.3921 \pm 5.1013i$	0.8119	7.6628
	VC ₁₁	$-0.1890 \pm 3.1510i$	0.5015	5.9859
	VC ₁₂	$-0.6579 \pm 7.2393i$	1.1522	9.0509
	$\Delta\omega_{12}$	$-0.4580 \pm 6.0078i$	0.9562	7.6014
	$\Delta\omega_{13}$	$-0.3249 \pm 5.1406i$	0.8182	6.3069
	$\Delta\omega_{14}$	$-0.2198 \pm 3.1981i$	0.5090	6.8575
	$\Delta\omega_{15}$	$-0.3723 \pm 4.8889i$	0.7781	7.5933
	$\Delta\omega_{16}$	$-0.2013 \pm 3.1881i$	0.5074	6.3019

These four WADCs were implemented in the ANATEM software to perform time-domain simulations. A three-phase fault with a duration of 100 ms was applied to bus 40 in the four operating cases described in Table 1 of this test system. Figures 3, 4, 5 and 6 show the angular response of generator 14 after the occurrence of this temporary contingency for cases OC0, OC1, OC2 and OC3 respectively for the different WADCs designed by the CSA, PSO, SCA and MSCA techniques. Figures 7, 8, 9 and 10 already show the angular responses of this generator 14 for this same contingency for cases OC0, OC1, OC2 and OC3 respectively and loss of the communication channel of the WADC control signal that will go to the AVR of generator 11. Figures 11, 12, 13 and 14 already show the angular responses of this generator 14 for this same contingency for cases OC0, OC1, OC2 and OC3 respectively and loss of the speed signal from generator 16. In addition to the angle signals from the generators, the output control signals from the WADC can also be evaluated after the contingency occurs, both when the WADC is operating fully and when there is a communication failure. Figures 15, 16, 17 and 18 show the VC_{11} output signal from the WADC after the occurrence of this temporary contingency for cases OC0, OC1, OC2 and OC3 respectively. Figures 19, 20, 21 and 22 show the VC_{11} output signal from the WADC for this same contingency for cases OC0, OC1, OC2 and OC3 respectively and loss of the communication channel of the WADC control signal that will go to the AVR of generator 11. Figures 23, 24, 25 and 26 already show the VC_{11} output signal from the WADC for this same contingency for cases OC0, OC1, OC2 and OC3 respectively and loss of the speed signal from generator 16. These results allow the following discussions:

- Angular responses are significantly affected by permanent communication channel losses of the different WADCs. Angular responses can be quite oscillatory if the closed-loop control system does not have high damping ratios.
- In all cases, the system with the WADC-MSCA designed by the proposed method provided the best damped angular responses, a desired scenario to ensure small-signal angular stability in power systems.
- The proposed method based on linearized models and eigenvalue analysis ensured that the loss of a communication channel did not adversely affect the angular responses.
- The VC_{11} output signals from the WADC show a significant performance during and after the elimination of the three-phase fault.
- Figures 19, 20, 21 and 22 show a constant VC_{11} signal equal to 0 because the communication channel corresponding to the VC_{11} signal was permanently lost.

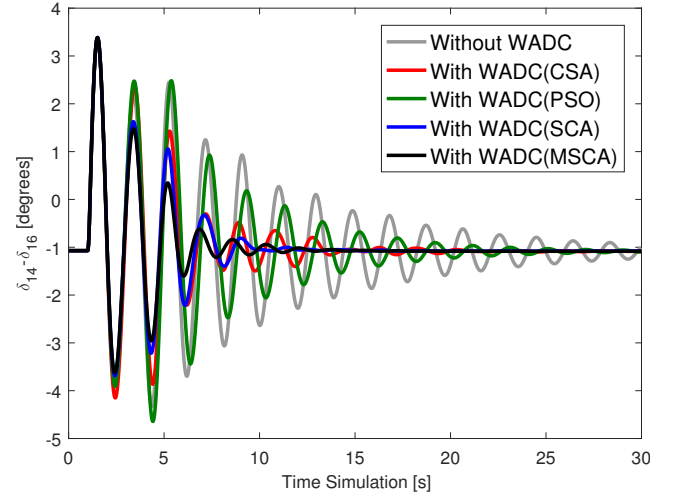


FIGURE 3. Generator angle 14 for case OC0.

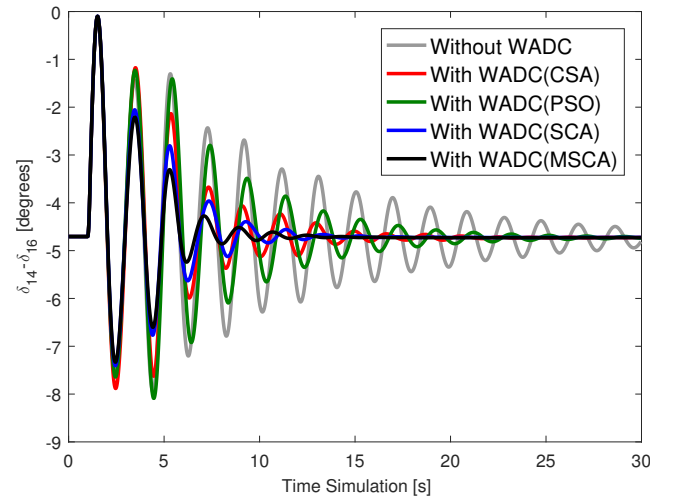


FIGURE 4. Generator angle 14 for case OC1.

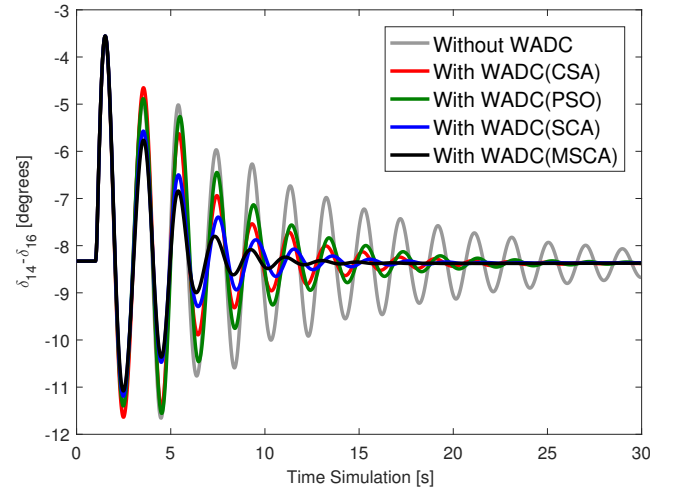


FIGURE 5. Generator angle 14 for case OC2.

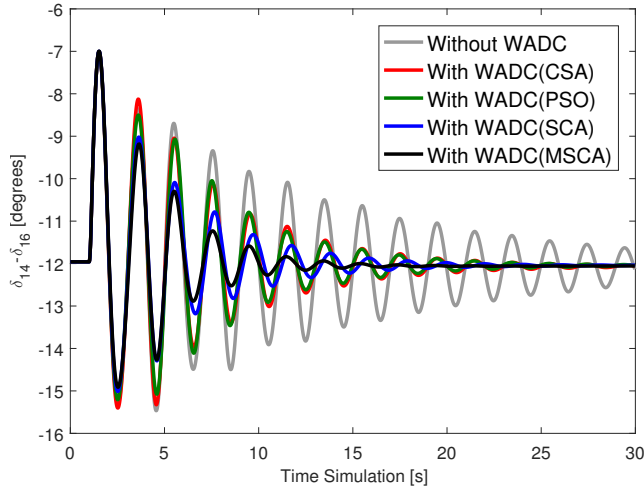


FIGURE 6. Generator angle 14 for case OC3.

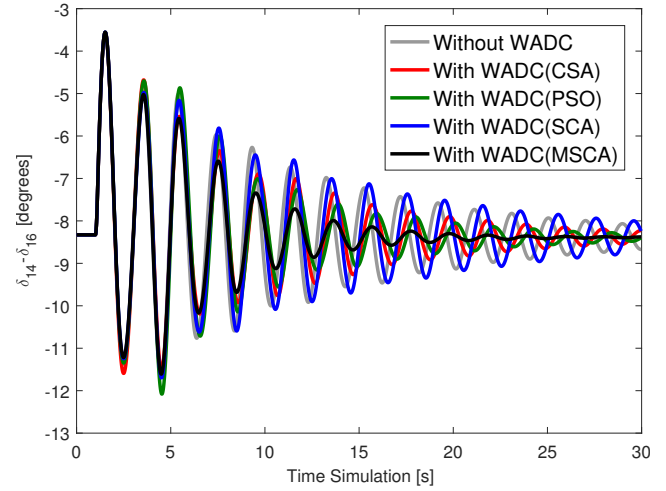


FIGURE 9. Generator angle 14 for OC2 case and permanent loss of generator AVR control signal 11.

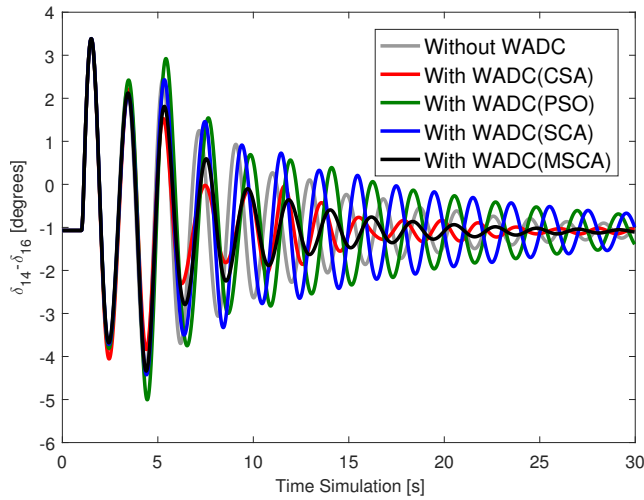


FIGURE 7. Generator angle 14 for OC0 case and permanent loss of generator AVR control signal 11.

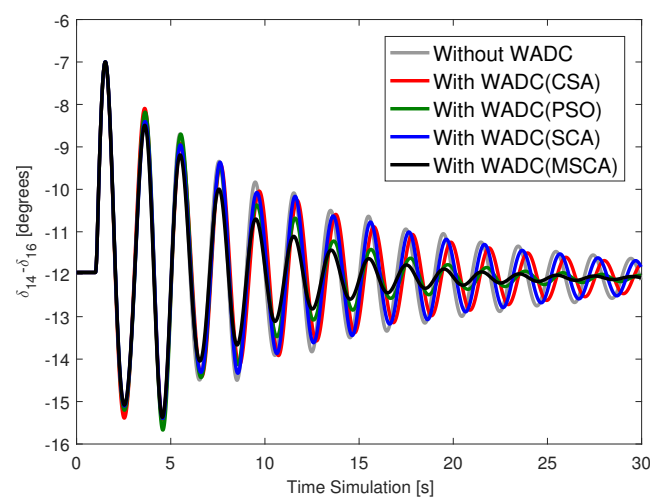


FIGURE 10. Generator angle 14 for OC3 case and permanent loss of generator AVR control signal 11.

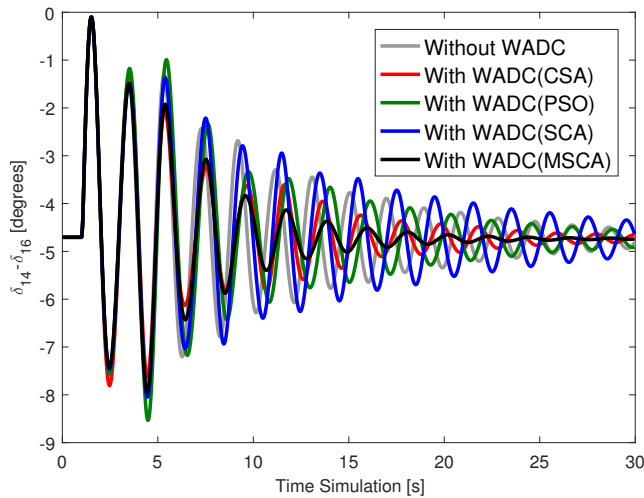


FIGURE 8. Generator angle 14 for OC1 case and permanent loss of generator AVR control signal 11.

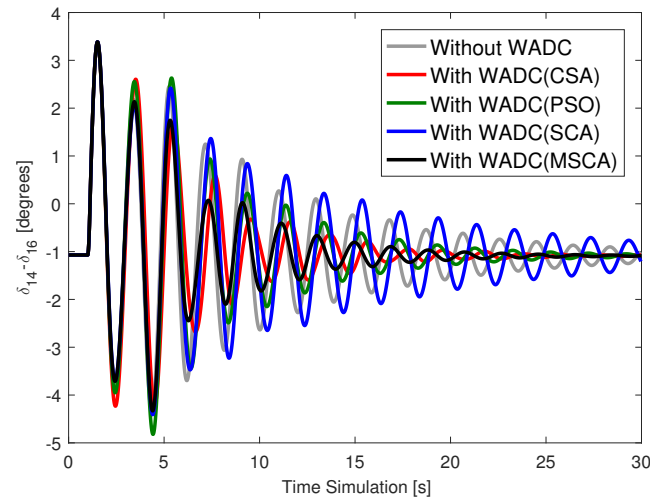


FIGURE 11. Generator angle 14 for case OC0 and permanent loss of generator speed signal 16

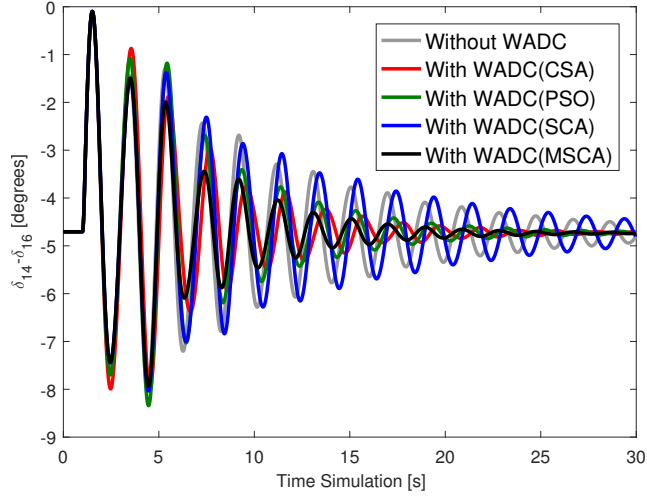


FIGURE 12. Generator angle 14 for case OC1 and permanent loss of generator speed signal 16

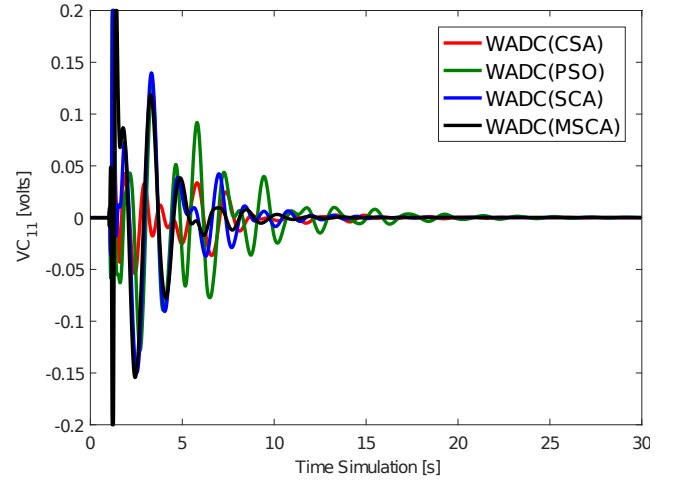


FIGURE 15. VC_{11} output signal from the WADC for case OC0.

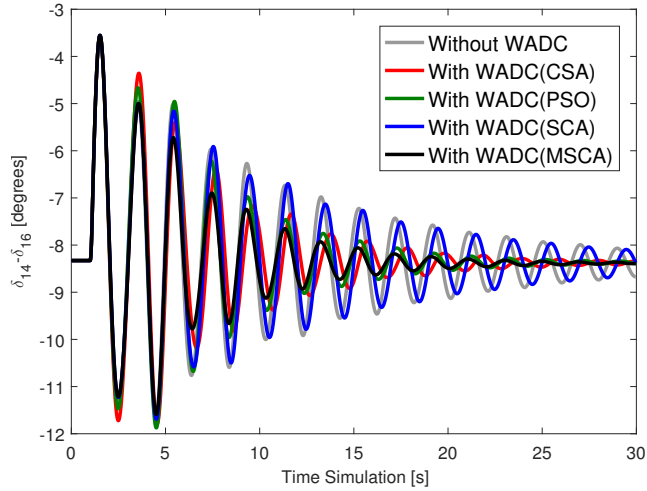


FIGURE 13. Generator angle 14 for case OC2 and permanent loss of generator speed signal 16

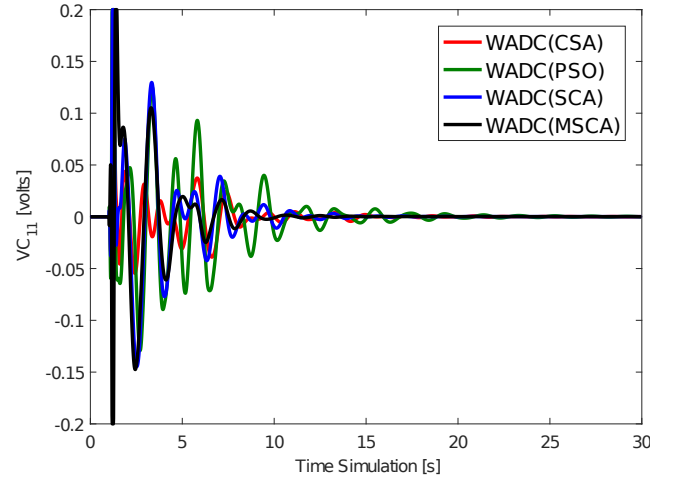


FIGURE 16. VC_{11} output signal from the WADC for case OC1.

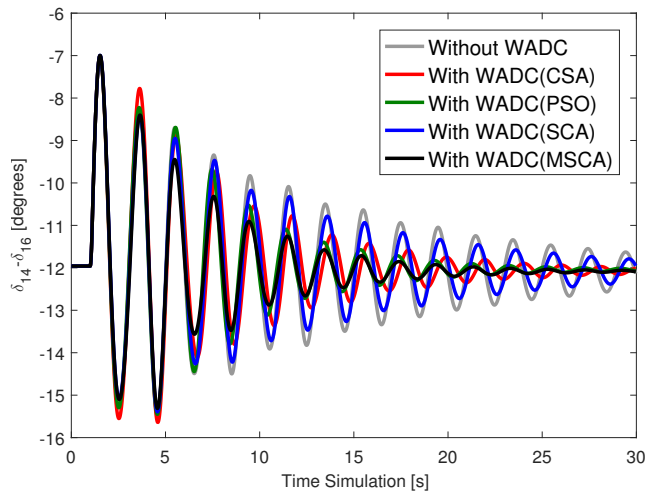


FIGURE 14. Generator angle 14 for case OC3 and permanent loss of generator speed signal 16

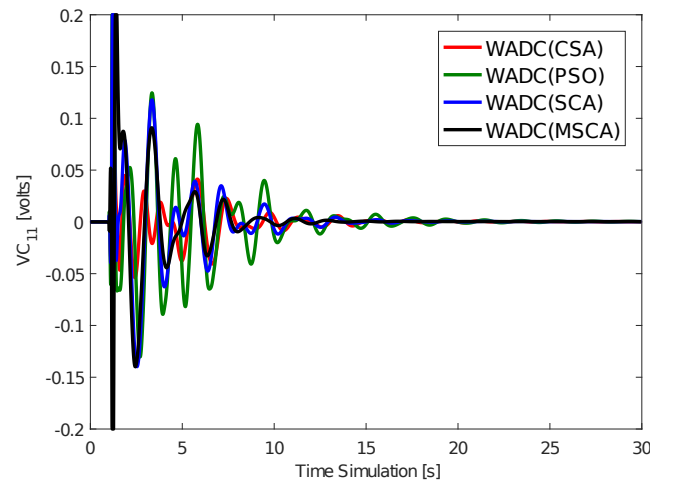


FIGURE 17. VC_{11} output signal from the WADC for case OC2.

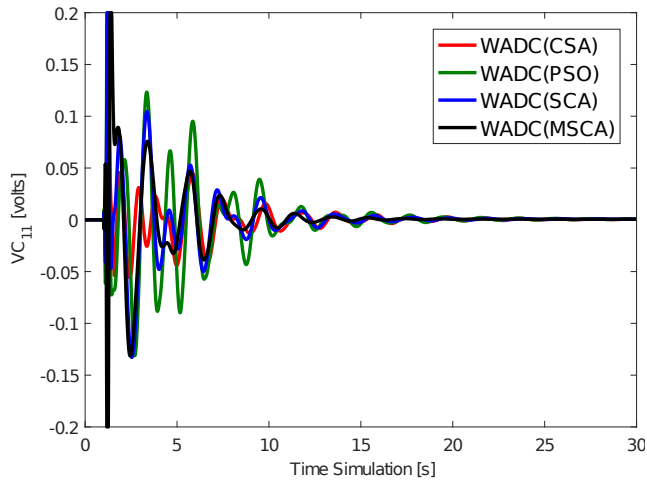


FIGURE 18. VC_{11} output signal from the WADC for case OC3.

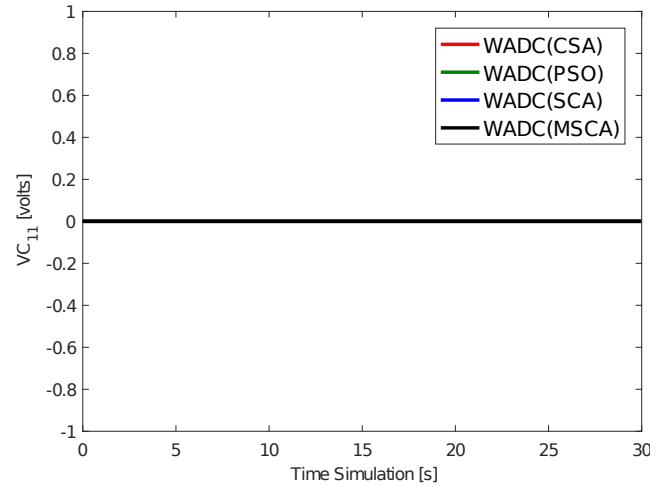


FIGURE 21. VC_{11} output signal from the WADC for OC2 case and permanent loss of generator AVR control signal 11.

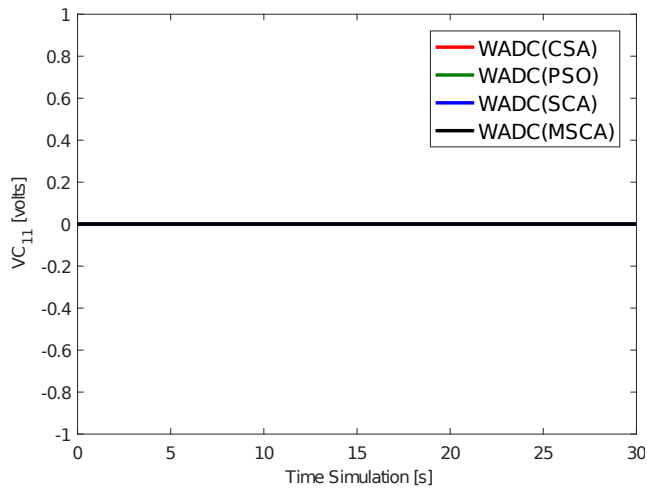


FIGURE 19. VC_{11} output signal from the WADC for OC0 case and permanent loss of generator AVR control signal 11.

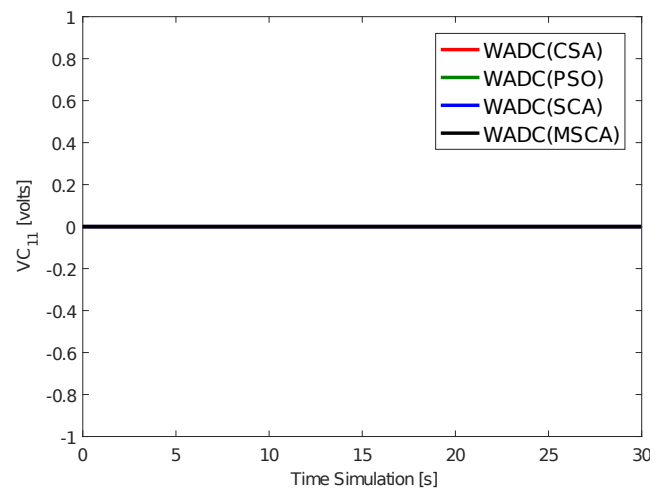


FIGURE 22. VC_{11} output signal from the WADC for OC3 case and permanent loss of generator AVR control signal 11.

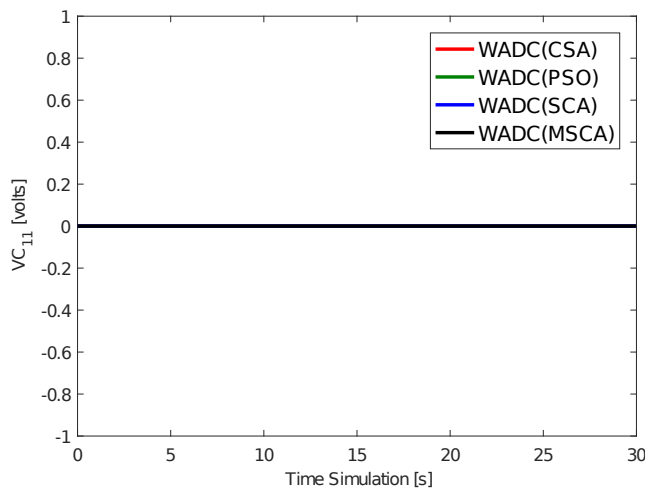


FIGURE 20. VC_{11} output signal from the WADC for OC1 case and permanent loss of generator AVR control signal 11.

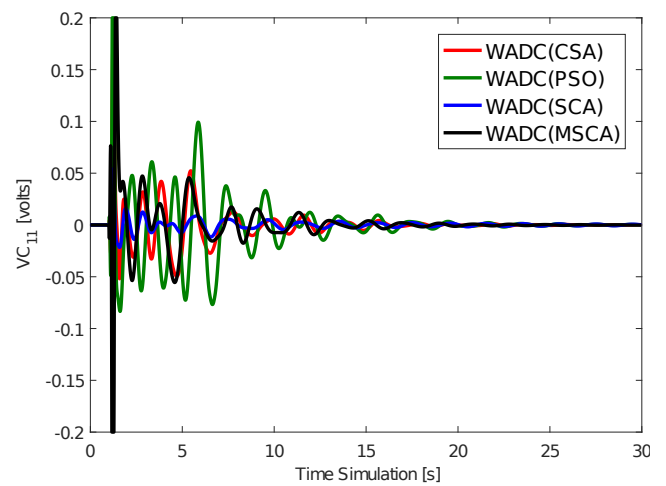


FIGURE 23. VC_{11} output signal from the WADC for case OC0 and permanent loss of generator speed signal 16

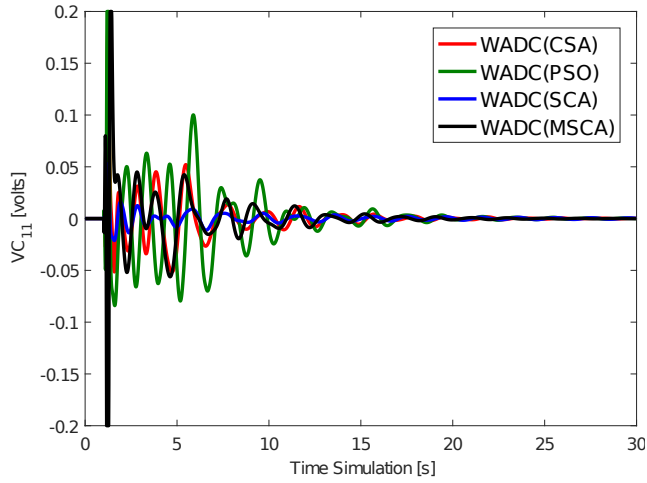


FIGURE 24. VC_{11} output signal from the WADC for case OC1 and permanent loss of generator speed signal 16

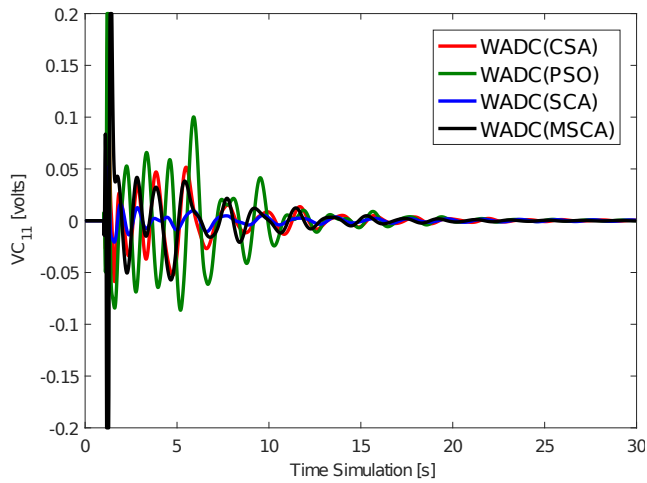


FIGURE 25. VC_{11} output signal from the WADC for case OC2 and permanent loss of generator speed signal 16

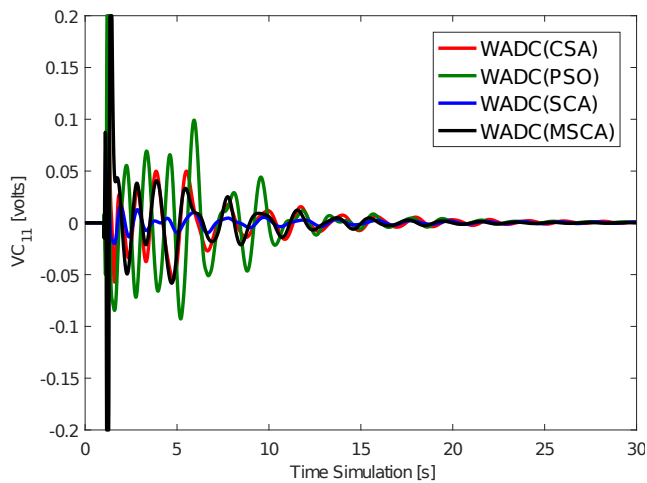


FIGURE 26. Generator angle 14 for case OC3 and permanent loss of generator speed signal 16

VI. CONCLUSIONS

This work proposes a method based on the modified sine cosine algorithm (MSCA) for the design of WADCs resilient to communication failures. The project consists of solving the multiobjective optimization model up to a limited number of iterations.

The case studies showed that MSCA converged to the lowest objective function values for the 100 different simulations when compared to the SCA, PSO, and CSA algorithms. Convergence to the lowest values ensured that the damping ratios of the system's oscillation modes were greater than 11% for the WADC operating with integrity and greater than 6% for the WADC operating with the loss of a communication channel.

Time-domain analyses with contingency applications showed that the angular responses of the system equipped with the WADC designed using the proposed method (WADC-MSCA) provided less oscillation and good damping for all operating scenarios, even when a communication channel was lost. Well-damped angular responses are one of the requirements for ensuring small-signal angular stability.

Future research will evaluate strategies to ensure the resilience of the WADC to multiple communication failures. Additionally, new strategies may be developed to combat cyberattacks such as False Data Injection Attacks. Time delay compensators will also be evaluated and developed to handle variable time delays. The WADC poles will be assessed as design variables in future research and how this may benefit control objectives.

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HENRIQUE R. DE ALMEIDA received the B.Sc. degree in electrical engineering from the Federal University of Rio de Janeiro (UFRJ), Rio de Janeiro, Brazil, in 2023.



MURILO E. C. BENTO (Member, IEEE) received the B.Sc., M.Sc., and Ph.D degrees in electrical engineering from the University of São Paulo (USP), São Carlos, Brazil, in 2013, 2016, and 2020, respectively.

He is currently an Adjunct Professor with the Federal University of Rio de Janeiro (UFRJ), Rio de Janeiro, Brazil. His research focuses on power system stability analysis and control, with emphasis on small-signal and voltage stability problems.

He is an Associate Editor of the journal *MDPI Electricity* and the journal *Springer Electrical Engineering*.

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